

Appendix A: Variable Definitions

This appendix lists every variable used in the empirical analyses, with construction formula, data source, and originating reference. Variable names follow paper-standard conventions where available (Dzieliński et al. 2021 for speech measures, Bates et al. 2009 for cash-holdings controls); deviations from source-paper frequency or sample are documented in the formula column.

A.1 Sample Manifest

Name	Description / Formula	Source	Reference
file_name, ceo_id, ceo_name, gvkey, ff12_code, ff12_name, start_date	Master sample manifest with CEO, firm, and industry identifiers	outputs/1.4_AssembleManifest	

A.2 Text/Linguistic Variables

Name	Description / Formula	Source	Reference
NegCall	Entire-call negative sentiment percentage. <i>Formula:</i> Ratio of negative words to total words in the entire call, based on LM 2011 negative-words list. Per DWZ 2021 Appendix Table A.1: 'percentage of negative words in all words spoken by the CEO, CFO and analysts attending the call.'	outputs/2_Textual_Analysis/2.1_NegCall	Dzieliński et al. (2021) (Section 4.4, p.18; Appendix Table A.1, p.54)
UncAnsCEO	CEO-only Q&A uncertainty percentage. <i>Formula:</i> CEO Q&A uncertainty (%) = $\text{UnctWordsAnsCEO} / \text{WordsAnsCEO}$ (DWZ 2021 Eqn 2). Uncertainty wordlist = LM 2011 Master Dictionary 'uncertainty' list (297 words, Aug 2014 version). CEO identified via Execucomp ceoann field.	outputs/2_Textual_Analysis/2.2_UncAnsCEO	Dzieliński et al. (2021) (Section 4.2, Eqn 2, p.13; Appendix Table A.1, p.54)
UncAnsMgr	Manager Q&A uncertainty percentage (all managers including CEO). <i>Formula:</i> Manager Q&A uncertainty (%) = $(\text{count of LM 2011 uncertainty-wordlist words spoken by managers in Q\&A segment}) / (\text{total words spoken by managers in Q\&A segment})$. Construction strategy follows DWZ 2021 Eqn (2); manager-pool identification follows BGT 2018 segment-split code (iangow/bgt replication). 'Manager' pool includes CEO plus other managerial speakers, extending DWZ which separates CEO/CFO.	outputs/2_Textual_Analysis/2.3_UncAnsMgr	Dzieliński et al. (2021) (measurement strategy, Eqn 2); Bushee et al. (2018) (manager-pool identification via iangow/bgt); thesis extension to pooled manager speakers
UncPreCEO	CEO-only presentation uncertainty percentage. <i>Formula:</i> CEO Presentation uncertainty (%) = $\text{UnctWordsPreCEO} / \text{WordsPreCEO}$ (DWZ 2021 Eqn 1). Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.4_UncPreCEO	Dzieliński et al. (2021) (Section 4.2, Eqn 1, p.13; Appendix Table A.1, p.54)

Name	Description / Formula	Source	Reference
UncPreMgr	Manager presentation uncertainty percentage. <i>Formula:</i> Manager Presentation uncertainty (%) = (count of LM 2011 uncertainty-wordlist words by managers in Presentation segment) / (total manager words in Presentation segment). Construction strategy follows DWZ 2021 Eqn (1); manager-pool identification and Pres/Ans segment split follow BGT 2018 via iangow/bgt replication code.	outputs/2_Textual_Analysis/FinVars	Dechow et al. (2021) (measurement strategy, Eqn 1); Bushee et al. (2018) (segment split + manager pool); thesis extension to pooled manager speakers
UncQue	Analyst Q&A uncertainty percentage. <i>Formula:</i> Analyst Q&A uncertainty (%) = UnctWordsQue / WordsQue (DWZ 2021 Eqn 3). Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/FinVars	Dechow et al. (2021) (Section 4.2, Eqn 3, p.13; Appendix Table A.1, p.54)

A.3 Financial / Market Variables

Name	Description / Formula	Source	Reference
BGTAvg_Amihud	BGT 25-day symmetric average Amihud illiquidity (H7e). <i>Formula:</i> Mean daily Amihud illiquidity over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). F1D extension to BGT 2018.	src/flid/shared/variables/bgt.long.window.amihud (BGTLongWindowAmihudBuilder, 3-col output)	Dechow et al. (2021) (symmetric average shape)
BGTAvg_Spread	BGT 25-day symmetric average bid-ask spread (H14e). <i>Formula:</i> Mean daily relative bid-ask spread over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). Spread = (ASK - BID) / ((ASK + BID) / 2). F1D extension to BGT 2018.	src/flid/shared/variables/bgt.long.window.spread (BGTLongWindowSpreadBuilder, 3-col output)	Dechow et al. (2021) (symmetric average shape)
BGTDelta_Amihud	BGT 25-day post-pre Amihud illiquidity change (H7d). <i>Formula:</i> Mean Amihud over post window [+1, +25] minus mean over pre window [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	src/flid/shared/variables/bgt.long.window.amihud (BGTLongWindowAmihudBuilder, 3-col output)	Dechow et al. (2021) (delta shape)
BGTDelta_Spread	BGT 25-day post-pre bid-ask spread change (H14d). <i>Formula:</i> Mean spread over [+1, +25] minus mean over [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	src/flid/shared/variables/bgt.long.window.spread (BGTLongWindowSpreadBuilder, 3-col output)	Dechow et al. (2021) (delta shape)
BGTLevel_Amihud	Compute BGT (2018) long-window Amihud illiquidity around earnings calls. <i>Formula:</i> CRSP via get_raw_daily_data (see module: src/flid/shared/variables/bgt.long.window.amihud.py)	CRSP via get_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) 56(1):85-121
BGTLevel_Spread	Compute BGT (2018) long-window closing bid-ask spread around earnings calls. <i>Formula:</i> CRSP via get_raw_daily_data (closing BID/ASK) (see module: src/flid/shared/variables/bgt.long.window.spread.py)	CRSP via get_raw_daily_data (closing BID/ASK)	Bushee, Gow & Taylor (2018, JAR) [window]; Lee (2016, TAR) [formula]
Capex	Build Capex = capxy_annual / atq_lag from raw Compustat quarterly data. <i>Formula:</i> Compustat/capxy_Q4/atq (see module: src/flid/shared/variables/capex.intensity.py)	Compustat/capxy_Q4/atq	Bates, Kahle & Stulz (2009, JF)

Name	Description / Formula	Source	Reference
CashFlowAt	Build CashFlow = oancfy / atq from raw Compustat quarterly data. <i>Formula:</i> Compustat/oancfy/atq (see module: src/fl1d/shared/variables/cash_flow.py)	Compustat/oancfy/atq	Bates, Kahle & Stulz (2009, JF)
CashRatio	Build CashRatio = cheq / atq from raw Compustat quarterly data. <i>Formula:</i> Compustat/cheq/atq (see module: src/fl1d/shared/variables/cash_holdings.py)	Compustat/cheq/atq	Bates et al. (2009) (Section II, p.1991; BKS use annual che/at — we use quarterly cheq/atq for panel alignment)
ChangDebtChoice	ChangDebtChoice (description not in module docstring) <i>Formula:</i> see module: src/fl1d/shared/variables/_compustat_engine.py	src/fl1d/shared/variables/_compustat_engine.py	Chang (2006)
ChangExternalFund	ChangExternalFunding (description not in module docstring) <i>Formula:</i> see module: src/fl1d/shared/variables/_compustat_engine.py	src/fl1d/shared/variables/_compustat_engine.py	Chang (2006)
DailyVola	Build Volatility from raw CRSP daily stock files via CRSPEngine. <i>Formula:</i> CRSP/RET (see module: src/fl1d/shared/variables/volatility.py)	CRSP/RET	dwz2021
DebtToCapital	Build DebtToCapital = total debt / (equity + total debt). <i>Formula:</i> Compustat/dlcq,dlttq,seqq (see module: src/fl1d/shared/variables/debt_to_capital.py)	Compustat/dlcq,dlttq,seqq	1995
DivDummy	Build DivDummy = (dvy > 0).astype(float) from raw Compustat quarterly data. <i>Formula:</i> Compustat/dvy>0 (see module: src/fl1d/shared/variables/dividend_payer.py)	Compustat/dvy>0	Bates et al. (2009) (Section IV, p.2000; dividend payout dummy based on common dividends)
DivPayerQ	Build DivPayerQ = (dvpsxq > 0).astype(float) from raw Compustat quarterly data. <i>Formula:</i> Compustat/dvpsxq>0 (quarterly ex-date) (see module: src/fl1d/shared/variables/dividend_payer_quarterly.py)	Compustat/dvpsxq>0 (quarterly ex-date)	Hoberg & Prabhala (2009, RFS) — Compustat item 26 (dvpsx) analog
EarnVol	Builder for Earnings Volatility (earnings_volatility) variable. <i>Formula:</i> Compustat (see module: src/fl1d/shared/variables/earnings_volatility.py)	Compustat	Duong et al. (2025) (Appendix B, p.30; 3-year rolling std of qoq asset-scaled earnings)
EquityDelayCon	Hoberg-Maksimovic (2015) equity-market financing constraint index (H22). <i>Formula:</i> Hoberg-Maksimovic (2015) firm-year equity-market financing constraint measure. Higher values = more constrained firm. Panel column is lowercase ‘equitydelaycon’; Camel-Case name is the spec convention. See docs/provenance/h22.md for merge details.	inputs/Hoberg_Maksimovic2015 (pre-computed) merged to panel	2015
FirmMat	Builder for Firm Maturity (firm_maturity) variable. <i>Formula:</i> Compustat (see module: src/fl1d/shared/variables/firm_maturity.py)	Compustat	Biddle et al. (2009) (Appendix A, p.130; years since first CRSP appearance); alt Leary and Roberts (2010) (App. 7 A p.354)
Leverage	Build Leverage = (dlcq + dlttq) / atq (interest-bearing debt only). <i>Formula:</i> Compustat/dlcq,dlttq,atq (see module: src/fl1d/shared/variables/book_lev.py)	Compustat/dlcq,dlttq,atq	BKS2009, rz1995
lnAssets	Build Size = ln(atq) from raw Compustat quarterly data. <i>Formula:</i> Compustat/atq (see module: src/fl1d/shared/variables/size.py)	Compustat/atq	dwz2021

Name	Description / Formula	Source	Reference
Loss	Builder for Loss (H5 Control Variable). <i>Formula:</i> Compustat ibq ; 0 (see module: src/fl1d/shared/variables/loss_dummy.py)	Compustat ibq ; 0	Biddle et al. (2009) (Appendix A, p.131; indicator for negative net income before extraordinary items)
PayoutRatio_q	Build PayoutRatio_q = (dvpspq × cshoq) / ibq from Compustat. <i>Formula:</i> Compustat/(dvpspq*cshoq)/ibq (quarterly) (see module: src/fl1d/shared/variables/payout_ratio_quarterly.py)	Compustat/(dvpspq*cshoq)/ibq (quarterly)	hls2004
PRisk	Match quarterly PRisk onto each earnings call by (gvkey, calendar_quarter). <i>Formula:</i> Hassan et al. (2019) quarterly PRisk (see module: src/fl1d/shared/variables/prisk_q.py)	Hassan et al. (2019) quarterly PRisk	hhlt2019
RDSales	Build RDSales = xrdq / atq from raw Compustat quarterly data. <i>Formula:</i> Compustat/xrdq,atq (see module: src/fl1d/shared/variables/rd_intensity.py)	Compustat/xrdq,atq	bks2009, jjl2021
RepurchaseIntensity	Build RepurchaseIntensity = quarterly_prstkcy / lagged_atq from Compustat. <i>Formula:</i> Compustat/quarterly_prstkcy/lagged_atq (see module: src/fl1d/shared/variables/repurchase_intensity.py)	Compustat/quarterly_prstkcy/lagged_atq	Thesis/Abe et al. suggestion
ROA	Build ROA = iby_annual / avg_assets from Compustat quarterly data. <i>Formula:</i> Compustat/iby,atq (see module: src/fl1d/shared/variables/roa.py)	Compustat/iby,atq	Wang (2020), dwz2021
SalesGrowth	Build SalesGrowth from raw Compustat quarterly data (annual Q4 panel). <i>Formula:</i> Compustat/saley_Q4_YoY (see module: src/fl1d/shared/variables/sales_growth.py)	Compustat/saley_Q4_YoY	Biddle et al. (2009) (Section 3.2, p.117; pct change in sales)
sCFO	Build sCFO = rolling 5-year std (min 3 yrs) of (oancfy/atq_{t-1}) per gvkey. <i>Formula:</i> Compustat/oancfy/atq_{t-1} rolling std (see module: src/fl1d/shared/variables/ocf_volatility.py)	Compustat/oancfy/atq_{t-1} rolling std	Biddle et al. (2009) (Appendix A, p.130; 5-year rolling std of CFO/avg assets)
StockPrice	Compute stock price at earnings call date (VECTORIZED). <i>Formula:</i> CRSP via get_raw_daily_data (see module: src/fl1d/shared/variables/stock_price.py)	CRSP via get_raw_daily_data	Amiram et al. (2016) (Appendix A, p.137; daily CRSP stock price; justification cites Stoll 1978)
SurpDec	Signed-quintile earnings surprise rank (-5 to +5) within calendar quarter. <i>Formula:</i> Signed quintile rank of percentage earnings surprise per DWZ 2021 Appendix Table A.1: raw surprise = (actual - consensus forecast earnings) / share price 5 trading days before announcement, x 100. Firms grouped into 5 bins of positive surprise (+5 largest through +1 smallest positive), 0 for zero surprises, and 5 bins of negative surprise (-1 smallest negative through -5 largest negative). DWZ footnote 32 notes this decile convention follows Hirshleifer, Lim & Teoh (2009) and DellaVigna & Pollet (2009).	src/fl1d/shared/variables/surprise_rank (via IbesEngine)	DellaVigna & Pollet (2021) (Appendix Table A.1, p.55; convention cites DellaVigna-Pollet 2009 + Hirshleifer-Lim-Teoh 2009)
TobinsQ	Build TobinsQ = (AT + cshoq*prccq - CEQ) / AT from raw Compustat quarterly data. <i>Formula:</i> Compustat/(atq+cshoq*prccq-ceq)/atq (see module: src/fl1d/shared/variables/tobins_q.py)	Compustat/(atq+cshoq*prccq-ceq)/atq	Botosan, Kahle & Stulz (2009, JF)

Name	Description / Formula	Source	Reference
Turnover	Compute share turnover at earnings call date (VECTORIZED). <i>Formula:</i> CRSP via get_raw_daily_data (see module: src/fld/shared/variables/turnover.py)	CRSP via get_raw_daily_data	Amihud (2002) (Section 2.1, p.33; alt rolling-window def in Chang et al. (2006) Appendix B p.3045)

A.4 Econometric / Indicator Variables

Name	Description / Formula	Source	Reference
BelowIG	Below-IG credit rating subsample indicator (H1.2). <i>Formula:</i> Binary = 1 if firm's S&P long-term issuer credit rating is below investment grade (BB+ and below) at call date; 0 otherwise. Rated firms only (Unrated firms get separate indicator).	runtime (S&P credit rating subsample indicator)	thesis (H1.2 rating subsample)
HighTSIMM	High product-market-similarity indicator (H1.1). <i>Formula:</i> Binary = 1 if firm-year TotalSimilarity (HP 2016 product market similarity) is above the within-year median; 0 otherwise.	runtime (within-year median split of TotalSimilarity)	hp2016 (TSIMM construction); thesis (H1.1 split convention)
Lagged_DV	Generic lagged-DV control (1Q lag of suite-specific DV). <i>Formula:</i> 1-quarter lag of the dependent variable, constructed at runtime per spec (DV varies per suite). Always included as base control to absorb persistence.	runtime (runner-constructed via .shift(1) within gvkey)	thesis convention (per feed-back_lagged_dv.md)
Unrated	No-rating subsample indicator (H1.2). <i>Formula:</i> Binary = 1 if firm has no S&P rating at call date; 0 otherwise.	runtime (S&P credit rating subsample indicator)	thesis (H1.2 rating subsample)
z_log_TotalSimilarity	z-scored log of HP 2016 TotalSimilarity (continuous moderator, H1.1). <i>Formula:</i> z-score of $\log(1 + \text{TotalSimilarity})$ computed within year. TotalSimilarity = HP 2016 firm-year product market similarity score (sum of pairwise cosine similarities $\times 100$).	runtime (transformation of HP 2016 TSIMM)	hp2016

A.5 Runtime Derivatives (Lead/Lag/Centered/Interaction)

Name	Description / Formula	Source	Reference
AbsSurpDec	Absolute earnings surprise quintile magnitude. <i>Formula:</i> abs(SurpDec) — unsigned magnitude of signed-quintile earnings surprise rank, values 0-5.	runtime (runner computes —SurpDec—; see run_h14c/d/e)	thesis (magnitude control independent of sign)
BGTavg_Amihud_lead1Q	1Q lead of BGTavg_Amihud. <i>Formula:</i> 1-quarter lead of BGTavg_Amihud	runtime (runner .shift in gvkey group)	bgt2018 (window); F1D extension (symmetric average shape)
BGTavg_Spread_lead1Q	1Q lead of BGTavg_Spread. <i>Formula:</i> 1-quarter lead of BGTavg_Spread	runtime (runner .shift in gvkey group)	bgt2018 (window); F1D extension (symmetric average shape)
BGTDelta_Amihud_lead1Q	1Q lead of BGTDelta_Amihud. <i>Formula:</i> 1-quarter lead of BGTDelta_Amihud	runtime (runner .shift in gvkey group)	bgt2018 (window); F1D extension (delta shape)
BGTDelta_Spread_lead1Q	1Q lead of BGTDelta_Spread. <i>Formula:</i> 1-quarter lead of BGTDelta_Spread	runtime (runner .shift in gvkey group)	bgt2018 (window); F1D extension (delta shape)

Name	Description / Formula	Source	Reference
BGTLevel_Amihud_lead1	1Q lead of BGTLevel_Amihud. <i>Formula:</i> 1-quarter lead of BGTLevel_Amihud	runtime (runner .shift in gvkey group)	bgt2018
BGTLevel_Spread_lead1	1Q lead of BGTLevel_Spread. <i>Formula:</i> 1-quarter lead of BGTLevel_Spread	runtime (runner .shift in gvkey group)	bgt2018, lee2016
Capex_lead	1Q lead of Capex. <i>Formula:</i> 1-quarter lead of Capex	runtime (runner .shift in gvkey group)	Bates, Kahle & Stulz (2009, JF)
Capex_lead2	2Q lead of Capex. <i>Formula:</i> 2-quarter lead of Capex	runtime (runner .shift in gvkey group)	Bates, Kahle & Stulz (2009, JF)
Capex_lead3	3Q lead of Capex. <i>Formula:</i> 3-quarter lead of Capex	runtime (runner .shift in gvkey group)	Bates, Kahle & Stulz (2009, JF)
Capex_lead4	4Q lead of Capex. <i>Formula:</i> 4-quarter lead of Capex	runtime (runner .shift in gvkey group)	Bates, Kahle & Stulz (2009, JF)
CashRatio_lead	1Q lead of CashRatio. <i>Formula:</i> 1-quarter lead of CashRatio	runtime (runner .shift in gvkey group)	see base var
ChangDebtChoice_lead1	1Q lead of ChangDebtChoice. <i>Formula:</i> 1-quarter lead of ChangDebtChoice	runtime (runner .shift in gvkey group)	cdh2006
ChangExternalFunding_lead1	1Q lead of ChangExternalFunding. <i>Formula:</i> 1-quarter lead of ChangExternalFunding	runtime (runner .shift in gvkey group)	cdh2006
DebtToCapital_lead1	1Q lead of DebtToCapital. <i>Formula:</i> 1-quarter lead of DebtToCapital	runtime (runner .shift in gvkey group)	rz1995
DeltaILLIQ_lead1	1Q lead of DeltaILLIQ. <i>Formula:</i> 1-quarter lead of DeltaILLIQ	runtime (runner .shift in gvkey group)	amihud02
DISP_lag	1Q lag of DISP. <i>Formula:</i> 1-quarter lag of DISP	runtime (runner .shift in gvkey group)	Wang (2020)
DISP_lead	1Q lead of DISP. <i>Formula:</i> 1-quarter lead of DISP	runtime (runner .shift in gvkey group)	Wang (2020)
DivPayerQ_lead1	1Q lead of DivPayerQ. <i>Formula:</i> 1-quarter lead of DivPayerQ	runtime (runner .shift in gvkey group)	Hoberg & Prabhala (2009, RFS) – primary; Chetty & Saez (2005, QJE) – secondary (firm-quarter frequency precedent)
DSPREAD_lead1	1Q lead of DSPREAD. <i>Formula:</i> 1-quarter lead of DSPREAD	runtime (runner .shift in gvkey group)	lee2016
EquityDelayCon_lead1	1Q lead of EquityDelayCon. <i>Formula:</i> 1-quarter lead of EquityDelayCon	runtime (runner .shift in gvkey group)	hm2015
Leverage_lead	1Q lead of Leverage. <i>Formula:</i> 1-quarter lead of Leverage	runtime (runner .shift in gvkey group)	bks2009, rz1995
PayoutRatio_q_lead1	1Q lead of PayoutRatio_q (qtr-aligned). <i>Formula:</i> 1-quarter lead of PayoutRatio_q (calendar-quarter aligned)	runtime (runner .shift in gvkey group)	dds2004
PostCallAmihud	Panel-time post-call Amihud illiquidity level (H7b). <i>Formula:</i> Panel-time mean daily Amihud illiquidity over the post-call period (call quarter + N following quarters per H7b runner spec). Computed on panel structure with two-way clustering.	runtime (panel-time computation in H7b runner)	thesis (H7b panel-time post-call construction)

Name	Description / Formula	Source	Reference
PostCallAmihud_lead	1Q lead of PostCallAmihud. <i>Formula:</i> 1-quarter lead of PostCallAmihud	runtime (runner .shift in gvkey group)	thesis (H7b panel-time post-call construction)
PostCallSpread	Panel-time post-call bid-ask spread level (H14b). <i>Formula:</i> Panel-time mean daily relative bid-ask spread over the post-call period (call quarter + N following quarters per H14b runner spec). Computed on panel structure with two-way clustering.	runtime (panel-time computation in H14b runner)	thesis (H14b panel-time post-call construction)
PostCallSpread_lead	1Q lead of PostCallSpread. <i>Formula:</i> 1-quarter lead of PostCallSpread	runtime (runner .shift in gvkey group)	thesis (H14b panel-time post-call construction)
PRisk_lag	1Q lag of PRisk. <i>Formula:</i> 1-quarter lag of PRisk	runtime (runner .shift in gvkey group)	hhlt2019
PRisk_lag2	2Q lag of PRisk. <i>Formula:</i> 2-quarter lag of PRisk	runtime (runner .shift in gvkey group)	hhlt2019
RDSales_lead	1Q lead of RDSales. <i>Formula:</i> 1-quarter lead of RDSales	runtime (runner .shift in gvkey group)	bks2009, jjl2021
RepurchaseIntensity_lead	1Q lead of RepurchaseIntensity (qtr-aligned). <i>Formula:</i> 1-quarter lead of RepurchaseIntensity (calendar-quarter aligned)	runtime (runner .shift in gvkey group)	Thesis Advisor Suggestion
UncAnsMgr_c	Mean-centered UncAnsMgr. <i>Formula:</i> UncAnsMgr minus its sample mean (mean-centered for interaction interpretability)	runtime (sample mean subtracted)	thesis (interaction interpretability)
UncAnsMgr_c x BelowIG	Interaction term: UncAnsMgr_c x BelowIG. <i>Formula:</i> Product: UncAnsMgr_c * BelowIG	runtime (product of two predictors)	thesis (interaction term)
UncAnsMgr_c x HighTSIMM	Interaction term: UncAnsMgr_c x HighTSIMM. <i>Formula:</i> Product: UncAnsMgr_c * HighTSIMM	runtime (product of two predictors)	thesis (interaction term)
UncAnsMgr_c x IG	Interaction term: UncAnsMgr_c x IG. <i>Formula:</i> Product: UncAnsMgr_c * IG	runtime (product of two predictors)	thesis (interaction term)
UncAnsMgr_c x Unrated	Interaction term: UncAnsMgr_c x Unrated. <i>Formula:</i> Product: UncAnsMgr_c * Unrated	runtime (product of two predictors)	thesis (interaction term)
UncAnsMgr_c x zlogTSIMM	Interaction term: UncAnsMgr_c x zlogTSIMM. <i>Formula:</i> Product: UncAnsMgr_c * zlogTSIMM	runtime (product of two predictors)	thesis (interaction term)

A.9 Stage 9

Name	Description / Formula	Source	Reference
CCCL	SEC comment letter received (binary, forward-looking) <i>Formula:</i> 1 if SEC comment letter received in next quarter, 0 otherwise	inputs/EDGAR.Comments	edh2017s
DeltaILLIQ	Change in Amihud illiquidity around earnings call <i>Formula:</i> PostCallILLIQ - PreCallILLIQ over [+1,+3] vs [-3,-1] days	inputs/CRSP.DSF	amihud02
DISP	Analyst forecast dispersion (Wang 2020) <i>Formula:</i> SD(latest analyst EPS forecasts in T-31..T-1 window) / prccq_prior	inputs/tr_ibes	Wang (2020)
DSPREAD	Change in closing bid-ask spread around earnings call <i>Formula:</i> $\text{mean}(\text{RelSpread}[+1,+3]) - \text{mean}(\text{RelSpread}[-3,-1])$ where $\text{RelSpread} = 2 * (\text{ASK} - \text{BID}) / (\text{ASK} + \text{BID})$	inputs/CRSP.DSF	lee2016

Name	Description / Formula	Source	Reference
GEPU_log	Davis (2016, NBER WP 22740) Global Economic Policy Uncertainty (GEPU) index (log-transformed, current-\$ GDP weights) <i>Formula:</i> $\log(\text{GEPU_current})$ — GDP-weighted average of 16 country EPU indices, matched by calendar month of call start_date	inputs/EconomicPolicyUncertainty/Global	Davis (2016)
GPR_log	Caldara & Iacoviello (2022, AER) headline Geopolitical Risk index (log-transformed, 10-newspaper recent version) <i>Formula:</i> $\log(\text{GPR})$ — recent headline Geopolitical Risk index, matched by calendar month of call start_date	inputs/matteoiacoviello/GPR2Global	CPR2Global
SEC_Letters_fwd	SEC comment letter count (forward-looking) <i>Formula:</i> Count of SEC-originated letters (UP-LOAD type) in next calendar quarter	inputs/EDGAR_Comments	edh2017 (thesis extension to count)
TotalSimilarity	Hoberg-Phillips total product similarity <i>Formula:</i> Sum of pairwise cosine similarities from 10-K product descriptions	inputs/TNIC	hp2016
US_EPU_log	Baker, Bloom & Davis (2016, QJE) US News-Based Economic Policy Uncertainty index (log-transformed) <i>Formula:</i> $\log(\text{News_Based_Policy_Uncert_Index})$ matched by calendar month of call start_date	inputs/EconomicPolicyUncertainty/US	bp2016

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